Sets: Questions and Full Solutions

Question 01

English

Consider the sets $P = \{x \in \mathbb{R} : x^3 = x\}$, $Q = \{x \in \mathbb{R} : |x| = x^2\}$, $R = \{x \in \mathbb{Z} : |x-1|+|x+1| = 2\}$, $T = \{x \in \mathbb{R} : |x-1| + |x+1| = 2\}$ and $W = \{x \in \mathbb{R} : |x-1| + |x+1| < 2\}$. Then which of the following statements is TRUE?

(A) $P = Q$ and $W = \{0\}$
(B) $Q \neq R$ and $W = \emptyset$
(C) $P = Q = R$ and $W = \emptyset$
(D) R and T are equivalent sets

Only one option is correct.

हिन्दी

समुच्चयों $P = \{x \in \mathbb{R} : x^3 = x\}$, $Q = \{x \in \mathbb{R} : |x| = x^2\}$, $R = \{x \in \mathbb{Z} : |x-1| + |x+1| = 2\}$, $T = \{x \in \mathbb{R} : |x-1| + |x+1| = 2\}$ तथा $W = \{x \in \mathbb{R} : |x-1| + |x+1| < 2\}$ पर विचार कीजिए। तब निम्न में से कौन-सा कथन सत्य है?

(A) $P = Q$ तथा $W = \{0\}$
(B) $Q \neq R$ तथा $W = \emptyset$
(C) $P = Q = R$ तथा $W = \emptyset$
(D) R तथा T तुल्य समुच्चय हैं

केवल एक विकल्प सही है।

Solution

Set P: $x^3 = x$ gives $x^3 - x = x(x-1)(x+1) = 0$, so $x \in \{-1, 0, 1\}$. Hence $P = \{-1, 0, 1\}$.

Set Q: since $x^2 = |x|^2$, the condition $|x| = x^2$ becomes $|x|^2 - |x| = 0$, that is, $|x|(|x| - 1) = 0$. So $|x| = 0$ or $|x| = 1$, giving $x \in \{-1, 0, 1\}$. Hence $Q = \{-1, 0, 1\}$.

For the sets R, T and W use the triangle inequality: $|x - 1| + |x + 1| = |1 - x| + |x + 1| \geq |(1 - x) + (x + 1)| = 2$ for every real x.

Equality holds if and only if $1 - x$ and $x + 1$ are both non-negative, that is, $-1 \leq x \leq 1$. (Directly: for $x > 1$ the sum is $2x > 2$ and for $x < -1$ it is $-2x > 2$.)

Therefore $T = \{x \in \mathbb{R} : -1 \leq x \leq 1\} = [-1, 1]$, an infinite set, and R, which keeps only the integers of T, is $R = \{-1, 0, 1\}$.

Since the sum is never less than 2, no real number satisfies $|x - 1| + |x + 1| < 2$; so $W = \emptyset$.

Statement (A): $P = Q$ is true, but at $x = 0$ the sum is $1 + 1 = 2$, which is not less than 2, so $0 \notin W$ and $W \neq \{0\}$. So (A) is wrong.

Statement (B): $Q = \{-1, 0, 1\} = R$, so $Q \neq R$ is false and (B) is wrong.

Statement (C): $P = Q = R = \{-1, 0, 1\}$ and $W = \emptyset$; both parts hold, so (C) is correct.

Statement (D): equivalent sets have the same number of elements. R is finite with $n(R) = 3$, while $T = [-1, 1]$ is infinite, so R and T are not equivalent and (D) is wrong.

Correct Option: (C)

Question 02

English

Let $A = \{x \in \mathbb{Z} : x^2 \leq 6x\}$ and $B = \{x \in \mathbb{Z} : 4 \leq x^2 \leq 16\}$. For sets M and N, let $M \Delta N = (M \setminus N) \cup (N \setminus M)$. Then the number of subsets X of $A \cup B$ such that X and A are equivalent sets, $X \neq A$ and $A \Delta X \subseteq B$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $A = \{x \in \mathbb{Z} : x^2 \leq 6x\}$ तथा $B = \{x \in \mathbb{Z} : 4 \leq x^2 \leq 16\}$ है। समुच्चयों M तथा N के लिए माना $M \Delta N = (M \setminus N) \cup (N \setminus M)$ है। तब $A \cup B$ के ऐसे उपसमुच्चयों X की संख्या, जिनके लिए X तथा A तुल्य समुच्चय हैं, $X \neq A$ है तथा $A \Delta X \subseteq B$ है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Set A: $x^2 \leq 6x \iff x(x-6) \leq 0 \iff 0 \leq x \leq 6$. Hence $A = \{0, 1, 2, 3, 4, 5, 6\}$ and $n(A) = 7$.

Set B: $4 \leq x^2 \leq 16 \iff 2 \leq |x| \leq 4$. Hence $B = \{-4, -3, -2, 2, 3, 4\}$ and $n(B) = 6$.

Then $A \cap B = \{2, 3, 4\}$, $A \setminus B = \{0, 1, 5, 6\}$, $B \setminus A = \{-4, -3, -2\}$ and $A \cup B = \{-4, -3, -2, 0, 1, 2, 3, 4, 5, 6\}$.

An element t lies in $A \Delta X$ exactly when it lies in one of A, X but not in the other. So $A \Delta X \subseteq B$ means: for every $t \notin B$, either t is in both A and X or in neither.

The elements of $A \cup B$ not in B are $0, 1, 5, 6$, all of which lie in A; hence each of them must lie in X. The elements of B are unrestricted.

So the admissible sets are exactly $X = \{0, 1, 5, 6\} \cup Y$ with $Y \subseteq B$, and different Y give different X.

X and A are equivalent iff $n(X) = n(A) = 7$, that is, $4 + n(Y) = 7$, $n(Y) = 3$. The number of 3-element subsets of the 6-element set B is $\binom{6}{3} = 20$.

Among these, $X = A$ occurs exactly once, for $Y = \{2, 3, 4\} = A \cap B$. The condition $X \neq A$ removes it.

Required number $= 20 - 1 = 19$.

Final Answer: 19

Question 03

English

Let S be the set of all ordered triples (x, y, z) of non-negative integers satisfying $x + y + z = 5$, and let $T = \{(x, y, z) \in S : xyz \neq 0\}$. Let m and n denote the numbers of elements of the power sets $\mathcal{P}(T)$ and $\mathcal{P}(S \setminus T)$, respectively. Then the value of $\log_4 \left(\frac{m}{n}\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना S उन सभी अऋणात्मक पूर्णांकों के क्रमित त्रिकों (x, y, z) का समुच्चय है जो $x + y + z = 5$ को संतुष्ट करते हैं, तथा माना $T = \{(x, y, z) \in S : xyz \neq 0\}$ है। माना m तथा n क्रमशः घात समुच्चयों $\mathcal{P}(T)$ तथा $\mathcal{P}(S \setminus T)$ के अवयवों की संख्याएँ हैं। तब $\log_4 \left(\frac{m}{n}\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

S is the set of non-negative integer solutions of $x + y + z = 5$. By the beggar's method (5 identical objects into 3 boxes, empty boxes allowed), $n(S) = \binom{5+3-1}{3-1} = \binom{7}{2} = 21$.

For $(x, y, z) \in S$, $xyz \neq 0$ holds iff none of x, y, z is 0, that is, $x, y, z \geq 1$.

Writing $x = 1 + x'$, $y = 1 + y'$, $z = 1 + z'$ with $x', y', z' \geq 0$ gives $x' + y' + z' = 2$, which has $\binom{2+2}{2} = \binom{4}{2} = 6$ non-negative solutions. So $n(T) = 6$ (the triples are the permutations of $(3, 1, 1)$ and $(2, 2, 1)$).

$T \subseteq S$, so $n(S \setminus T) = n(S) - n(T) = 21 - 6 = 15$ (the triples with at least one zero coordinate).

A set with k elements has 2^k subsets, so $m = n(\mathcal{P}(T)) = 2^6 = 64$ and $n = n(\mathcal{P}(S \setminus T)) = 2^{15}$.

Hence $\frac{m}{n} = 2^{6-15} = 2^{-9}$, and since $2 = 4^{1/2}$, $2^{-9} = 4^{-9/2}$.

Therefore $\log_4 \left(\frac{m}{n}\right) = -\frac{9}{2} = -4.5$.

Final Answer: -4.5

Question 04

English

Let $U = \{x \in \mathbb{Z} : |x| \leq 10\}$ be the universal set, and let $A = \{x \in U : x \text{ is a multiple of } 3\}$, $B = \{x \in U : x^2 > 20\}$ and $C = \{x \in U : x < 2\}$. For a subset Y of U, let $Y' = U \setminus Y$, and for subsets M, N of U, let $M \,\Delta\, N = (M \setminus N) \cup (N \setminus M)$. Let X be the subset of U satisfying $(A \,\Delta\, X)' = B \cup C'$. Then the sum of all the elements of X is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $U = \{x \in \mathbb{Z} : |x| \leq 10\}$ सार्वत्रिक समुच्चय है, तथा माना $B = \{x \in U : x^2 > 20\}$ तथा $C = \{x \in U : x < 2\}$ हैं, और A समुच्चय U के उन सभी अवयवों का समुच्चय है जो 3 के गुणज हैं। U के किसी उपसमुच्चय Y के लिए माना $Y' = U \setminus Y$ है, तथा U के उपसमुच्चयों M, N के लिए माना $M \,\Delta\, N = (M \setminus N) \cup (N \setminus M)$ है। माना X, U का वह उपसमुच्चय है जो $(A \,\Delta\, X)' = B \cup C'$ को संतुष्ट करता है। तब X के सभी अवयवों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Take complements (in U) of both sides: $A \,\Delta\, X = ((A \,\Delta\, X)')' = (B \cup C')'$.

By De Morgan's law, $(B \cup C')' = B' \cap (C')' = B' \cap C = C \setminus B$.

$B' = \{x \in U : x^2 \leq 20\} = \{-4, -3, \ldots, 4\}$ and $C = \{-10, \ldots, 1\}$, so $C \setminus B = B' \cap C = \{-4, -3, -2, -1, 0, 1\}$.

To recover X, use the algebraic properties of Δ: it is associative, $A \,\Delta\, A = \emptyset$ and $\emptyset \,\Delta\, X = X$. Hence $A \,\Delta\, (A \,\Delta\, X) = (A \,\Delta\, A) \,\Delta\, X = X$.

So $X = A \,\Delta\, (C \setminus B)$. Conversely this set satisfies the equation, so the subset X exists and is unique.

A consists of the multiples of 3 in U: $A = \{-9, -6, -3, 0, 3, 6, 9\}$.

$A \cap (C \setminus B) = \{-3, 0\}$, so $A \,\Delta\, (C \setminus B) = (A \cup (C \setminus B)) \setminus \{-3, 0\}$.

Hence $X = \{-9, -6, 3, 6, 9\} \cup \{-4, -2, -1, 1\} = \{-9, -6, -4, -2, -1, 1, 3, 6, 9\}$.

Sum of the elements $= (-9 - 6 + 3 + 6 + 9) + (-4 - 2 - 1 + 1) = 3 + (-6) = -3$. (Taking $A \cup (C \setminus B)$ instead of the symmetric difference would give -6.)

Final Answer: -3

Question 05

English

In a survey of 300 households of a town about three newspapers H, E and T, it was found that 150, 120 and 85 households read H, E and T, respectively; 45 households read both H and E, 30 read both E and T, and 20 read both T and H. The number of households that read none of the three newspapers is equal to the number of households that read all three. Among the households that read at least one of the three newspapers, exactly p percent read exactly two of them. Then the value of p is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

किसी नगर के 300 परिवारों में तीन समाचार पत्रों H, E तथा T के बारे में किए गए एक सर्वेक्षण में पाया गया कि 150, 120 तथा 85 परिवार क्रमशः H, E तथा T पढ़ते हैं; 45 परिवार H तथा E दोनों पढ़ते हैं, 30 परिवार E तथा T दोनों पढ़ते हैं, और 20 परिवार T तथा H दोनों पढ़ते हैं। तीनों में से कोई भी समाचार पत्र न पढ़ने वाले परिवारों की संख्या, तीनों समाचार पत्र पढ़ने वाले परिवारों की संख्या के बराबर है। तीनों में से कम से कम एक समाचार पत्र पढ़ने वाले परिवारों में से ठीक p प्रतिशत परिवार उनमें से ठीक दो समाचार पत्र पढ़ते हैं। तब p का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let k be the number of households that read all three newspapers, $k = n(H \cap E \cap T)$.

$n(H)+n(E)+n(T) = 150+120+85 = 355$ and $n(H\cap E)+n(E\cap T)+n(T\cap H) = 45+30+20 = 95$.

By the formula for three sets, $n(H \cup E \cup T) = 355 - 95 + k = 260 + k$.

The number reading none is $300 - n(H \cup E \cup T) = 40 - k$. This equals k, so $40 - k = k$, giving $k = 20$.

In the Venn diagram, a household in exactly two of the sets is counted once in the pairwise total 95, and a household in all three is counted three times. So exactly two $= 95 - 3k = 95 - 60 = 35$.

Check that the data are consistent (every region non-negative): $H \cap E$ only $= 25$, $E \cap T$ only $= 10$, $T \cap H$ only $= 0$; H only $= 150 - 25 - 0 - 20 = 105$, E only $= 120 - 25 - 10 - 20 = 65$, T only $= 85 - 10 - 0 - 20 = 55$; total in at least one $= 105 + 65 + 55 + 25 + 10 + 0 + 20 = 280 = 260 + k$.

So 280 households read at least one newspaper, and 35 of them read exactly two.

$$p = \frac{35}{280} \times 100 = \frac{100}{8} = 12.5.$$

Final Answer: 12.5

Question 06

English

Let A, B and C be three mutually independent events such that $P(A) = \frac{1}{2}$, the probability that at least one of A, B, C occurs is 0.85, and the probability that exactly one of A, B, C occurs is 0.4. Then the probability that exactly two of A, B, C occur is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना A, B तथा C तीन परस्पर स्वतंत्र घटनाएँ हैं, जिनके लिए $P(A) = \frac{1}{2}$ है, A, B, C में से कम से कम एक घटना के घटित होने की प्रायिकता 0.85 है, तथा A, B, C में से ठीक एक घटना के घटित होने की प्रायिकता 0.4 है। तब A, B, C में से ठीक दो घटनाओं के घटित होने की प्रायिकता _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $P(B) = b$ and $P(C) = c$. By independence, the complements A', B', C' are also independent.

At least one of the events occurs is the complement of none occurring: $1 - P(A' \cap B' \cap C') = 0.85$, so $P(A')P(B')P(C') = \frac{1}{2}(1-b)(1-c) = 0.15$, that is, $(1-b)(1-c) = 0.3$, or $b + c - bc = 0.7$.

Exactly one occurs: either A occurs and neither B nor C, or A fails and exactly one of B, C occurs.

So $P(\text{exactly one}) = \frac{1}{2}(1-b)(1-c) + \frac{1}{2}\left[b(1-c) + c(1-b)\right] = 0.15 + \frac{1}{2}(b + c - 2bc)$.

Setting this equal to 0.4: $\frac{1}{2}(b + c - 2bc) = 0.25$, so $b + c - 2bc = 0.5$.

Subtracting from $b + c - bc = 0.7$ gives $bc = 0.2$, and then $b + c = 0.9$; so $\{b, c\} = \{0.4, 0.5\}$ (roots of $t^2 - 0.9t + 0.2 = 0$), both valid probabilities.

All three occur: $P(A \cap B \cap C) = \frac{1}{2} \cdot bc = 0.1$.

Exactly two occur: $P(A)\left[b(1-c) + c(1-b)\right] + P(A')\,bc = \frac{1}{2}(0.5) + \frac{1}{2}(0.2) = 0.25 + 0.1 = 0.35$.

Check: none + exactly one + exactly two + all three $= 0.15 + 0.4 + 0.35 + 0.1 = 1$.

Final Answer: 0.35

Question 07

English

Let $S = \{1, 2, 3, \ldots, 10\}$. Then the number of non-empty subsets X of S such that the product of all the elements of X is divisible by 12 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $S = \{1, 2, 3, \ldots, 10\}$ है। तब S के ऐसे अरिक्त उपसमुच्चयों X की संख्या, जिनके सभी अवयवों का गुणनफल 12 से विभाज्य है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$12 = 2^2 \cdot 3$. The product of the elements of X is divisible by 12 iff it contains a factor 3 and at least two factors 2.

Factor 3: the elements of S divisible by 3 are $3, 6, 9$, so the condition is $X \cap \{3, 6, 9\} \neq \emptyset$.

Factors of 2: the even elements are $2, 6, 10$ (one factor 2 each), 4 (two) and 8 (three). The power of 2 in the product is at least 2 iff $X \cap \{4, 8\} \neq \emptyset$ or X contains at least two of $2, 6, 10$.

Count the complement among all $2^{10} = 1024$ subsets of S (including $\emptyset$). Let F = subsets with $X \cap \{3, 6, 9\} = \emptyset$ and E = subsets in which the power of 2 is at most 1.

$n(F)$: the elements $3, 6, 9$ are excluded and the other 7 are free, so $n(F) = 2^7 = 128$.

$n(E)$: 4 and 8 are excluded, at most one of $2, 6, 10$ is chosen ($1 + 3 = 4$ ways), and the odd elements $1, 3, 5, 7, 9$ are free: $n(E) = 4 \cdot 2^5 = 128$.

$n(E \cap F)$: $3, 6, 9, 4, 8$ are excluded, at most one of $2, 10$ is chosen ($1 + 2 = 3$ ways), and $1, 5, 7$ are free: $n(E \cap F) = 3 \cdot 2^3 = 24$.

By inclusion-exclusion, the number of subsets whose product is divisible by 12 is $1024 - (128 + 128 - 24) = 792$.

The empty subset (empty product 1) belongs to $E \cap F$, so it is not among these 792; all of them are non-empty. (Equivalently, with $2^{10} - 1 = 1023$ non-empty subsets: $1023 - 127 - 127 + 23 = 792$.)

Final Answer: 792